



PROFESSIONAL CERTIFICATION

Node.js Development Training

Master Node.js to build fast scalable and efficient server-side applications. Learn Express.js APIs databases authentication and deployment for real-world backend solutions.



30

Hours



Certified

Trainers



Industry

Expert



Professional Training | Industry-Recognized Certificate

Dubai, UAE

About Orbit Training Center

Your Gateway to Professional Excellence in Dubai



Who We Are

Established in 2023, Orbit Training Center is a leading professional training provider in Dubai, offering 60+ professional certification courses across diverse fields, including engineering, IT, design, and languages.



Our Benefits

-  Industry recognized certifications
-  Convenient location in the heart of Dubai
-  Affordable fees with multiple payment plans
-  Lifetime post-training support

★Our Features

-  Hands-on learning with real-world projects
-  Expert trainers with industry experience
-  Flexible learning options to fit your schedule
-  State-of-the-art infrastructure & facilities



Our Mission

We deliver outstanding courses, expert teaching, strong career guidance, and placement support to help every student achieve their best results. We stay connected with alumni through networking events, lifelong learning, and ongoing mentorship. We run our operations with smart systems and data-driven decisions to create a smooth, exceptional experience for all.

Node.js Development

Build powerful server-side applications with Node.js and Express.

Course Description

This comprehensive course teaches you to develop robust backend applications using Node.js. You will master core Node.js concepts Express.js framework RESTful APIs database integration with MongoDB user authentication security best practices and deployment strategies. Through practical projects gain hands-on experience creating high-performance

Duration

30

Hours of Training

Format

Classroom / Online

Flexible Option

Key Outcomes

- ✓ Set up Node.js environment and projects
- ✓ Build web servers with Express.js framework
- ✓ Create and consume RESTful APIs
- ✓ Integrate databases like MongoDB
- ✓ Implement user authentication and security
- ✓ Handle errors and optimize performance
- ✓ Deploy Node.js applications to cloud platforms

Target Audience

- Aspiring backend developers exploring Node.js
- Full-stack developers enhancing server skills
- Programmers transitioning to JavaScript backend

Certification

- Certificate of Course Completion
- Industry Recognized Certification
- Trained by Certified Professionals

Prerequisites

- Basic knowledge of JavaScript programming
- Understanding of web technologies and HTTP
- Familiarity with command line interface

Node.js Development

Start building professional server-side applications with step-by-step modules from Node.js fundamentals to advanced Express development and deployment.

01 JavaScript Fundamentals (Variables and Data Types)

- > var let and const (differences and when to use each)
- > Primitive data types string number boolean null and undefined
- > Type conversion and type coercion in JavaScript
- > Template literals and string interpolation with backticks
- > Comments code structure and best practices in JavaScript

02 JavaScript Operators Control Flow and Loops

- > Arithmetic comparison and logical operators in JavaScript
- > if else and switch statements for conditional logic
- > for while and do-while loops for repeating code
- > break and continue for controlling loop execution
- > Ternary operator for short conditional expressions

03 JavaScript Functions and Scope

- > Declaring functions with function keyword and expressions
- > Arrow functions (syntax and differences from regular functions)
- > Function parameters default values and rest parameters
- > Understanding scope (global function and block scope)
- > Closures (functions that remember their outer scope)

04 JavaScript Arrays and Objects

- > Creating and accessing arrays and common array methods
- > map filter reduce forEach and find on arrays
- > Creating objects properties and accessing with dot notation
- > Destructuring arrays and objects for cleaner code
- > Spread and rest operators with arrays and objects

05 JavaScript ES6 and Modern Features

- > Modules — import and export for splitting code into files
- > Promises — creating and handling async operations
- > async and await for cleaner asynchronous code
- > Error handling with try catch and finally blocks
- > Optional chaining and nullish coalescing operators

06 JavaScript DOM and Browser Basics

- > Selecting and manipulating HTML elements with JavaScript
- > Handling click submit and keyboard events on elements
- > Fetch API (making HTTP requests and handling responses)
- > JSON (parsing and stringifying data for APIs)
- > Local storage (saving and reading data in the browser)

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07 Introduction to Node.js and Server-Side JavaScript

- > What Node.js is and how it differs from browser JavaScript
- > Installing Node.js and running JavaScript files with node
- > The Node.js event loop and non-blocking I/O explained
- > Using the Node.js REPL for quick code testing
- > Node.js global objects (__dirname __filename process module)

08 Node.js Core Modules and File System

- > Using the fs module to read write and delete files
- > Working with the path module for file path operations
- > Using the os module to get system information
- > Creating and using the http module for basic server
- > Using the events module and EventEmitter for custom events

09 NPM and Package Management

- > What NPM is and how the Node.js ecosystem works
- > Initializing a project with npm init and package.json
- > Installing local and global packages with npm install
- > Understanding dependencies and devDependencies in package.json
- > Using nodemon for automatic server restarts during development

10 Express.js (Building Web Servers)

- > What Express.js is and why it is used with Node.js
- > Creating an Express app and setting up a basic server
- > Defining routes with GET POST PUT and DELETE methods
- > Using route parameters and query strings in Express
- > Sending JSON responses and setting HTTP status codes

11 Express Middleware and Request Handling

- > What middleware is and how it works in Express
- > Using built-in middleware (express.json and express.static)
- > Writing custom middleware for logging and validation
- > Handling form data and file uploads in Express
- > Error handling middleware for catching and responding to errors

12 Working with Databases (MongoDB and Mongoose)

- > Introduction to MongoDB and NoSQL document databases
- > Connecting Node.js to MongoDB using Mongoose
- > Defining Mongoose schemas and models for collections
- > CRUD operations (create find update and delete documents)
- > Data validation and required fields in Mongoose schemas

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13 RESTful API Design with Express and MongoDB

- > What a REST API is and how it works with HTTP methods
- > Building a full REST API with Express routes and controllers
- > Organizing code with MVC (models views and controllers)
- > Using Postman to test API endpoints during development
- > Handling errors and sending consistent API responses

15 Environment Variables and Project Configuration

- > What environment variables are and why they are used
- > Using dotenv to manage .env files in Node.js projects
- > Storing sensitive data (DB URI secret keys and ports)
- > Switching between development and production configurations
- > Git best practices (adding .env to .gitignore correctly)

14 User Authentication with JWT

- > What authentication is and why it is needed in APIs
- > Hashing passwords securely using bcrypt library
- > Generating and signing JSON Web Tokens with jsonwebtoken
- > Protecting routes with JWT middleware authentication
- > Login and register flow — building auth endpoints in Express

16 Building and Deploying a Full Node.js Project

- > Planning and building a complete REST API from scratch
- > Connecting all layers (routes controllers models and auth)
- > Serving a frontend HTML CSS JS from Express static folder
- > Preparing app for deployment (environment and port setup)
- > Deploying Node.js app to a live server using Render or Railway

Final Projects and Assessment:

- **Assessment:** Project-based evaluation through real-world projects to assess practical skills and application without theory exams.
- **Tailored Curriculum:** Customized to industry standards, covering essential topics for professional development in relevant fields.
- **Hands-on Projects:** Real-world simulations and interactive tasks to build applicable skills and create professional portfolios.
- **Certification Prep:** Targeted review sessions, practice exercises, and strategies for achieving industry certifications.
- **Ongoing Support:** Access to recorded materials, forums, instructor guidance, and career assistance for long-term growth.

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Muppuli Raja



One of great institute for REVIT BIM course. I completed my training with Mr Ahmed. He is one great trainer in this institute. He has very good knowledge for BIM Training. I completed my REVIT BIM training with orbit and it was very nice learning journey for me. And one of best thing is training center is very flexible for classes.



Arun



Best SolidWorks training in Dubai! The instructor made 3D modeling and assembly design easy to understand. The course covered all essential tools I need for product design and engineering work. Best training center for solidworks training. Best Training Center for Best Skills



Harshu CH



My experience at Orbit training center has been the best In every step I learn from there very clearly almost complete my course it's very interesting the trainer is the best professional and friendly ,and the rest of the team keep it up, thanks for supporting me Orbit and all staff...



Ranjan Axalan



I just finished my course in 3ds max. I learned a lot on how to create 3d model. Thank you for the experience and for the teaching method it help me to understand easily and grow my knowledge. I will use my learnings to my future company. Highly Recomanded!



Numan



I had good support from trainer Mr Ahmed and the other staff. They would schedule my BIM classes as per my convenience and would retake classes if I required. They always answer my calls when I need assistance. Not only that the trainer would go a mile a ahead by helping me with my work related issues like a true mentor.



Lelanie Fernandez



Overall, I highly recommend the Fusion 360 training at Orbit Center, especially with Ahmad as the instructor. This course has exceeded my expectations, and I am learning a lot. I would rate it 5 out of 5 stars!

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Aman Bakhtiyar



I had a great learning experience at Orbit Training Center. The trainers are knowledgeable, supportive, and always ready to help. The clear and practical teaching methods really helped improve my skills and confidence. I highly recommend Orbit Training Center to anyone looking to learn AutoCAD, Revit, and other professional skills.



Sana Zahid -



I recently completed the online Digital Marketing course with Orbit Training Center, and it was a great experience! The lessons were clear, practical, and easy to follow. The trainers were knowledgeable and supportive throughout the course. I've learned a lot that I can apply to my work here in Belgium. Highly recommend Orbit Training Center to anyone looking to improve their digital marketing skills!



Cleavon Henderson



Orbit Training Center offers outstanding CAD classes that are easy to understand, even as a first-year mechanical engineering student. The instructors simplify complex concepts and use practical examples, making the learning process engaging and effective. Whether you're new to CAD or improving your skills, Orbit Training Center provides a supportive and enriching environment. Highly recommended!



Danish Khan



Just finished Primavera P6 training at Orbit Training Center and it was fantastic! The trainer made complex scheduling concepts easy to understand with real-world examples. Got hands-on experience with project planning, resource allocation, and progress tracking. The course was well-structured and I feel confident using Primavera P6 for my projects now. Great value and would definitely recommend to other project managers!



500+ Students Trained

Join our growing community of professionals

95%

Satisfaction Rate

60+

Courses Offered

4.6

Average Rating



GET STARTED TODAY

Get in Touch with Us

At Orbit Training Center, we offer a diverse range of professional courses designed to equip you with the latest industry skills and knowledge. Our comprehensive programs cover in-demand fields such as Interior Design, Computer Science, Graphic Design, Digital Marketing, Solar Energy, and specialized software training including STAAD Pro, Revit Architecture, Tekla Structures, and more. Each course is crafted by industry experts to provide hands-on learning, real-world applications, and certification preparation.

Up to 30%

Early Bird Discount

Free

Demo Session



Call Us Now

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